

Individual Assignment 5

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Set up

Packages

```
#general use
library(here)

library(tidyverse)

library(janitor)

library(snakecase)

# packages for visualization customization

library(scales)
library(paletteer)
library(ggthemes)

# reading in .xlsx files
library(readxl)

library(NatParksPalettes)

# arranging plots
library(patchwork)
```

Data

```
# taxonomic information
taxon_list <- read_csv(here("data", "taxon_list.csv"))

# aquatic invertebrates
aq_ins <- read_xlsx(here("data", "Aquatic Sampling Data-2026-03-10.xlsx"),
  sheet = "Aquatic Insects")

# site information
sites <- read_xlsx(here("data", "Aquatic Sampling Data-2026-03-10.xlsx"),
  sheet = "sites")

# water quality information
water_quality <- read_xlsx(here("data", "Aquatic Sampling
  ↪ Data-2026-03-10.xlsx"),
  sheet = "Water Quality",
  na = c("", "n/a", "N/A", "over", "173+"))
```

copied from class

```
# creating new object from sites
ncos_sites <- sites |>

  # filter to only include NCOS quarterly sampling sites
  filter(sector == "NCOS" & sampling_frequency == "Quarterly")

water_quality_clean <- water_quality |>

  # cleaning column names
  clean_names() |>

  # filtering join: only include sites in ncos_sites
  # keep all the rows on the left that match up with rows on the right
  semi_join(ncos_sites, by = "site") |>

  # combine date and site together and remove those indiv. columns
  unite("date_site", date, site, remove = TRUE) |>

  # group by date_site
  group_by(date_site) |>
```

```

# create new columns that show median of each variable at each date_site
# because there are multiple samples at the same site at different depths
summarize(med_ph = median(p_h, na.rm = TRUE),
          med_sal = median(salinity_ppt, na.rm = TRUE),
          med_do = median(dissolved_oxygen_mg_l, na.rm = TRUE)) |>

# ungroup
ungroup() |>

# filter out the salinity outlier
filter(date_site != "2022-08-12_NVBR")

```

```

# creating new object from taxon_list
taxon_list_clean <- taxon_list |>

# converted taxon name values to snake case (lowercase)
mutate(verbatim_name = to_any_case(verbatim_name,
                                   case = "snake"))

```

```

# creating new clean aquatic invertebrates object from aquatic invertebrates
aq_ins_clean <- aq_ins |>

# cleaning column names
clean_names() |>

# filtering join: only include quarterly NCOS surveys from ncos_sites
semi_join(ncos_sites, by = "site") |>

# renaming column to just date
rename(date = date_on_vial) |>

# select columns of interest
select(date, sample_type, site,

       # 9 most abundant taxon
       ostracod,
       copepod,
       hemiptera_corixidae_boatman,
       cladocera,
       nematode,
       diptera_ceratopogonidae,

```

```

    annelida_oligochaete,
    diptera_chironomid,
    annelida_polychaete) |>

# filter to only include filtered beaker samples
filter(sample_type %in% c("FB250", "FB 250")) |>

pivot_longer(
  # cols = columns to make longer
  cols = ostracod:annelida_polychaete,
  # new column for the old column names
  names_to = "taxon",
  # new column for the values in each of the old columns
  values_to = "abundance"
) |>

# group by the date, site, and taxon
group_by(date, site, taxon) |>

# calculate average abundance per liter (sum abundance divided by 7.5
  ↳ liters)
summarize(avg_per_lit = sum(abundance, na.rm = TRUE)/7.5) |>

# ungroup
ungroup() |>

# create a new column called `date_site` to join with water quality
# don't remove the original columns
unite("date_site", date, site, remove = FALSE) |>

# join with water quality clean
# wqc has `date_site` column as well
left_join(water_quality_clean, by = "date_site") |>

# join with taxonomic information from taxon list
# verbatim_name column in taxon_list_clean matches with taxon column in
  ↳ aq_ins_clean
left_join(taxon_list_clean, by = c("taxon" = "verbatim_name")) |>

# fill in full site names
mutate(site_full = case_when(
  site == "NVBR" ~ "North of Venoco Bridge",

```

```

site == "NEC" ~ "East Channel",
site == "NMC" ~ "Main Channel",
site == "NPB" ~ "South of Phelps Bridge",
site == "NPB1" ~ "North of Phelps Bridge",
site == "NPB2" ~ "Phelps Road",
site == "NWP" ~ "West Pond",
site == "NDC" ~ "Devereux Creek"
)) |>

# setting factor levels for full site name by median salinity
mutate(site_full = fct_reorder(site_full, med_sal, .fun = "median", .na_rm
↪   = TRUE),
       site_full = fct_rev(site_full))

```

```

salinity_plot <- ggplot(data = aq_ins_clean,
                        # x-axis: full site name
                        # y-axis: median salinity
                        mapping = aes(x = site_full,
                                      y = med_sal)) +
# points representing median salinity at each survey
geom_jitter(height = 0,
            shape = 21) +

stat_summary(geom = "point",
            fun = median,
            color = "red",
            size = 2,) +

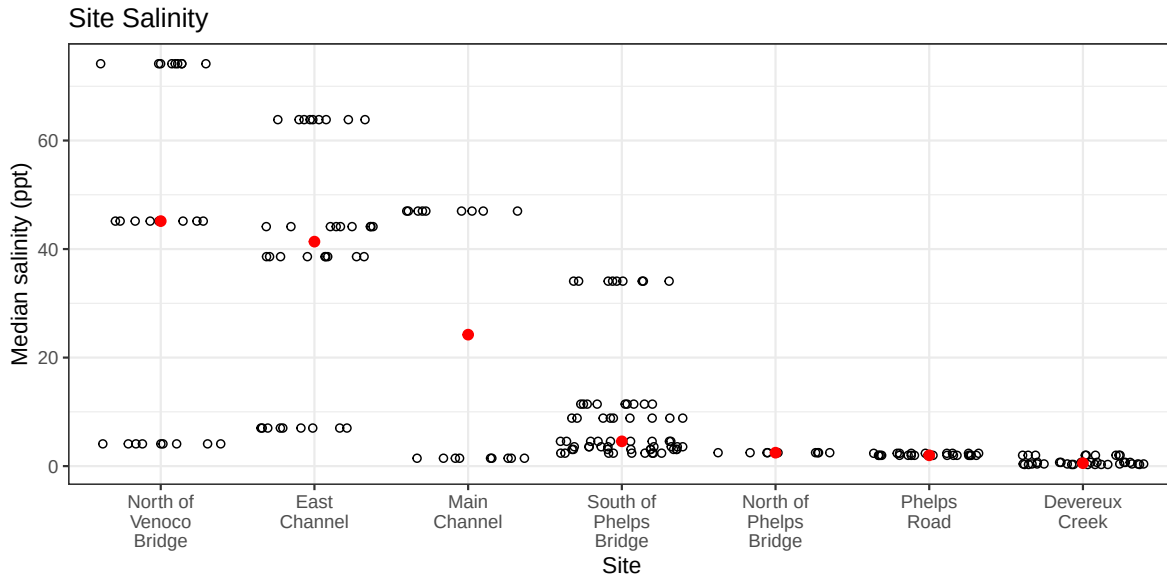
# wrap x-axis labels
scale_x_discrete(labels = label_wrap(width = 10)) +

# labeling titles and axes
labs(x = "Site",
     y = "Median salinity (ppt)",
     title = "Site Salinity") +

# change theme
theme_bw()

# displaying the object
salinity_plot

```



```
copepods <- aq_ins_clean |>

  filter(taxon == "copepod") |>

  mutate(log_avg_per_lit = log(avg_per_lit + 1))

# creating object to store plot
copepod_plot <- ggplot(data = copepods,
  # x-axis: full site name
  # y-axis: log transformed avg count/L
  mapping = aes(x = site_full,
    y = log_avg_per_lit)) +

  # jitter to represent salinity
  geom_jitter(height = 0,
    width = 0.1,
    shape = 21) +

  # points to represent median copepod abundance
  stat_summary(geom = "point",
    fun = median,
    color = "blue",
    size = 2) +

  # wrapping x axis labels
  scale_x_discrete(labels = label_wrap(width = 10)) +

  # relabelling axes
```

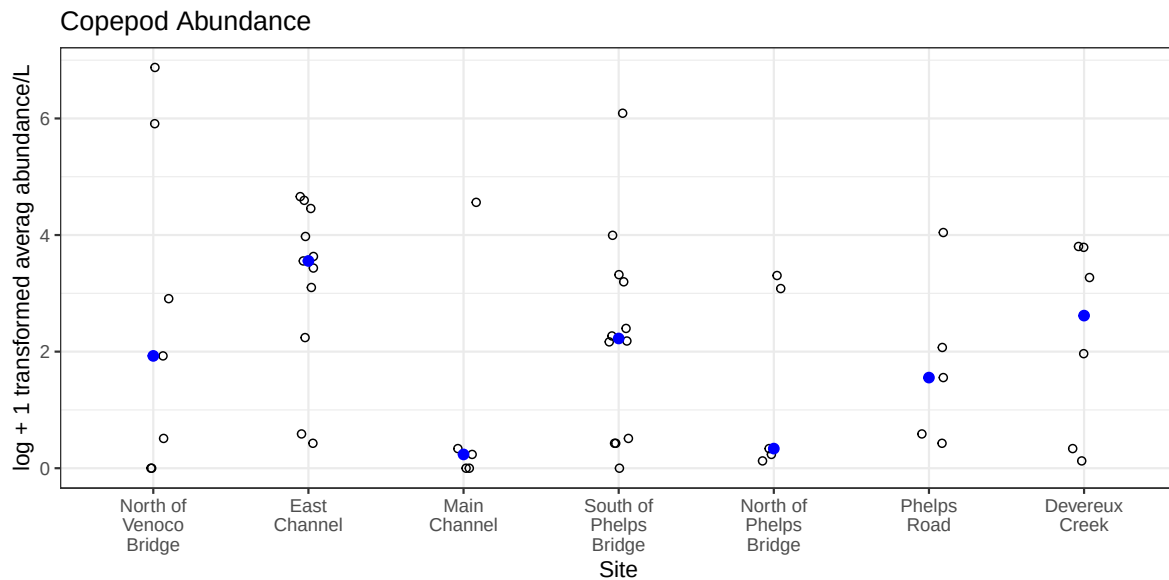
```

labs(x = "Site",
     y = "log + 1 transformed averag abundance/L",
     title = "Copepod Abundance") +

# different theme
theme_bw()

# display object
copepod_plot

```



1. Visualizing differences across sites in major taxon abundance

a. Plan your figure

- x-axis: site_full
- y-axis: average abundance of ostracod per liter
- geometry to represent average abundance/L: `geom_jitter()`
- geometry to represent medians: `stat_summary()`

b. Wrangle your data

```
# creating new object from aquatic insects
ostracod <- aq_ins_clean |>

# filter to only include ostracod
filter(taxon == "ostracod") |>

# calculate the average of ostracod abundance per lit
mutate(ostracod_log_avg_per_lit = log(avg_per_lit + 1))

# display 10 random rows

slice_sample(ostracod, n = 10)
```

```
# A tibble: 10 x 24
  date_site date site taxon avg_per_lit med_ph med_sal med_do
  <chr> <dtm> <chr> <chr> <dbl> <dbl> <dbl> <dbl>
1 2021-04-15~ 2021-04-15 00:00:00 NPB ostr~ 185. 8.09 3.57 8.24
2 2022-02-25~ 2022-02-25 00:00:00 NEC ostr~ 0 8.75 44.1 9.87
3 2022-11-19~ 2022-11-19 00:00:00 NDC ostr~ 6 NA NA NA
4 2023-09-15~ 2023-09-15 00:00:00 NMC ostr~ 0.4 NA NA 14.2
5 2023-03-03~ 2023-03-03 00:00:00 NDC ostr~ 5.73 7.45 0.688 5.64
6 2020-01-21~ 2020-01-21 00:00:00 NDC ostr~ 2.4 6.53 0.3 7.6
7 2022-11-18~ 2022-11-18 00:00:00 NVBR ostr~ 0 NA NA NA
8 2022-08-10~ 2022-08-10 00:00:00 NPB ostr~ 2.67 8.28 8.84 9.81
9 2021-11-12~ 2021-11-12 00:00:00 NPB ostr~ 79.2 NA NA NA
10 2023-06-06~ 2023-06-06 00:00:00 NPB1 ostr~ 0.533 7.48 NA 2.38
# i 16 more variables: taxonRank <chr>, scientificName <chr>, kingdom <chr>,
# Phylum <chr>, Class <chr>, Subclass <chr>, Superorder <chr>, Order <chr>,
# Infraorder <chr>, Family <chr>, Subfamily <chr>, Tribe <chr>, Genus <chr>,
# comments <chr>, site_full <fct>, ostracod_log_avg_per_lit <dbl>
```

c. Code your figure

```
# base layer: create an object that stores the plot
ostracod_plot <- ggplot(data = ostracod,
  # x = site
  # y = log + 1 transformed average abundance/L
```



```

        mapping = aes(x = site_full,
                      y = ostracod_log_avg_per_lit)) +

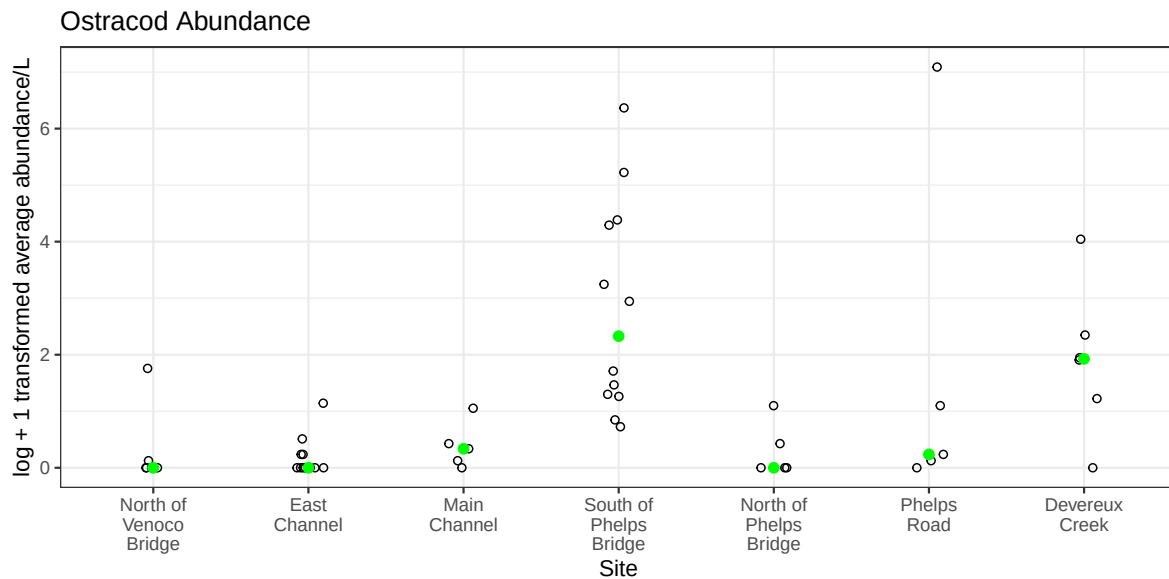
# first layer: geometry jitter points to represent each sample
geom_jitter(height = 0,
            width = 0.1,
            shape = 21) +
# points to represent median abundance
stat_summary(geom = "point",
            fun = median,
            color = "green",
            size = 2) +
# wrapping x axis labels
scale_x_discrete(labels = label_wrap(width = 10)) +

# labeling axes and title
labs(title = "Ostracod Abundance",
     x = "Site",
     y = "log + 1 transformed average abundance/L") +

# changed theme
theme_bw()

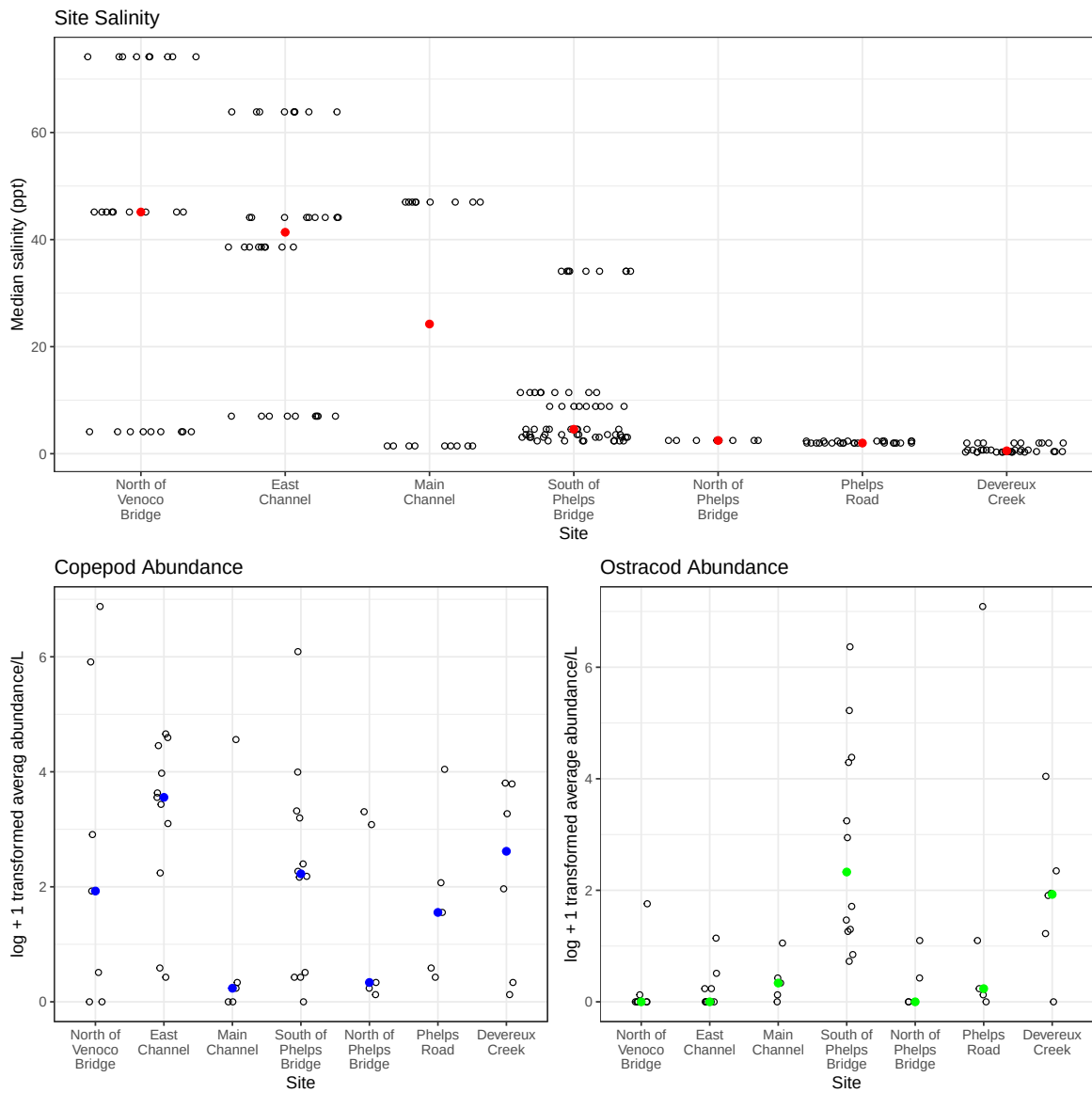
# display plot
ostracod_plot

```



d. Create a multipanel figure

```
salinity_plot / (copepod_plot | ostracod_plot)
```



e. Interpret your figure

What are the differences between sites in copepod and ostracod abundance?

- Average copepod abundance/L was highest at East Channel and Devereux Creek at respective transformed values of around 4 and 3, and moderate at a value of around 2 at North of Venoco Bridge, South of Phelps Bridge, and Phelps Road. The highest average ostracod abundance/L was moderate around a transformed value of 2 at South of Phelps Bridge and also Devereux Creek, and low at pretty much every other site.

How does ostracod abundance relate (or not relate) to site-level differences in salinity?

- Ostracod abundance does not correlate to salinity site-level differences in salinity because the plots do not show the same type of trend for the salinity plot and for the ostracod abundance plot.

2. Visualizing total abundance as a function of salinity

a. Plan your figure

- x-axis: salinity (ppt)
- y-axis: log-transformed average abundance/L
- color: taxon
- geometry: geom_jitter()
- facets: by each taxon in aq_ins_clean with free scales

b. Wrangle your data

```
# change existing aq_in_clean object
aq_ins_clean <- aq_ins_clean |>

# change taxon values to title case
mutate(taxon = str_replace_all(taxon, "_", " ") |> str_to_title()) |>

# group by the sample date and by taxon
group_by(date, taxon) |>

# add in a new column that is the sum of avg_per_lit per taxon per date
mutate(total_avg_per_lit = sum(avg_per_lit, na.rm = TRUE)) |>

# log transform the total abundance per lit
mutate(log_total_avg_per_lit = log(total_avg_per_lit + 1)) |>
```

```
# ungroup
ungroup() |>

# keep only date, taxon, med_sal, and total avg/L
select(date, taxon, med_sal, log_total_avg_per_lit) |>

# re-order taxon levels by max abundance
mutate(taxon = fct_reorder(taxon, log_total_avg_per_lit, max, .desc =
  ↪ TRUE))

# display 10 random rows
slice_sample(aq_ins_clean, n = 10)
```

```
# A tibble: 10 x 4
  date          taxon          med_sal log_total_avg_per_lit
  <dtm>         <fct>         <dbl>         <dbl>
1 2023-03-03 00:00:00 Nematode          1.98          0
2 2023-05-04 00:00:00 Cladocera          NA          0.125
3 2022-05-11 00:00:00 Nematode          0.398          0
4 2023-06-06 00:00:00 Copepod          NA          3.31
5 2023-09-12 00:00:00 Diptera Ceratopogonidae NA          0.427
6 2023-07-17 00:00:00 Annelida Polychaete NA          0
7 2023-07-17 00:00:00 Annelida Oligochaete NA          0
8 2023-11-12 00:00:00 Hemiptera Corixidae Boatman 47          0.125
9 2021-11-16 00:00:00 Nematode          2.38          0
10 2021-11-16 00:00:00 Annelida Polychaete 2.38          0
```

c. Code your figure

```
# base layer: create an object that stores the plot
abund_salinity_plot <- ggplot(data = aq_ins_clean,
  # x = salinity
  # y = log + 1 transformed average abundance/L
  mapping = aes(x = med_sal,
    y = log_total_avg_per_lit,
    color = taxon)) +

# first layer: geometry jitter points to represent each sample
geom_jitter(height = 0,
```

```

        width = 0.1,
        shape = 21) +

# facet by taxon
facet_wrap( ~taxon,
            scales = "free") +

# different color per taxon
scale_color_manual(values = natparks.pals("LakeNakuru", n = 9)) +

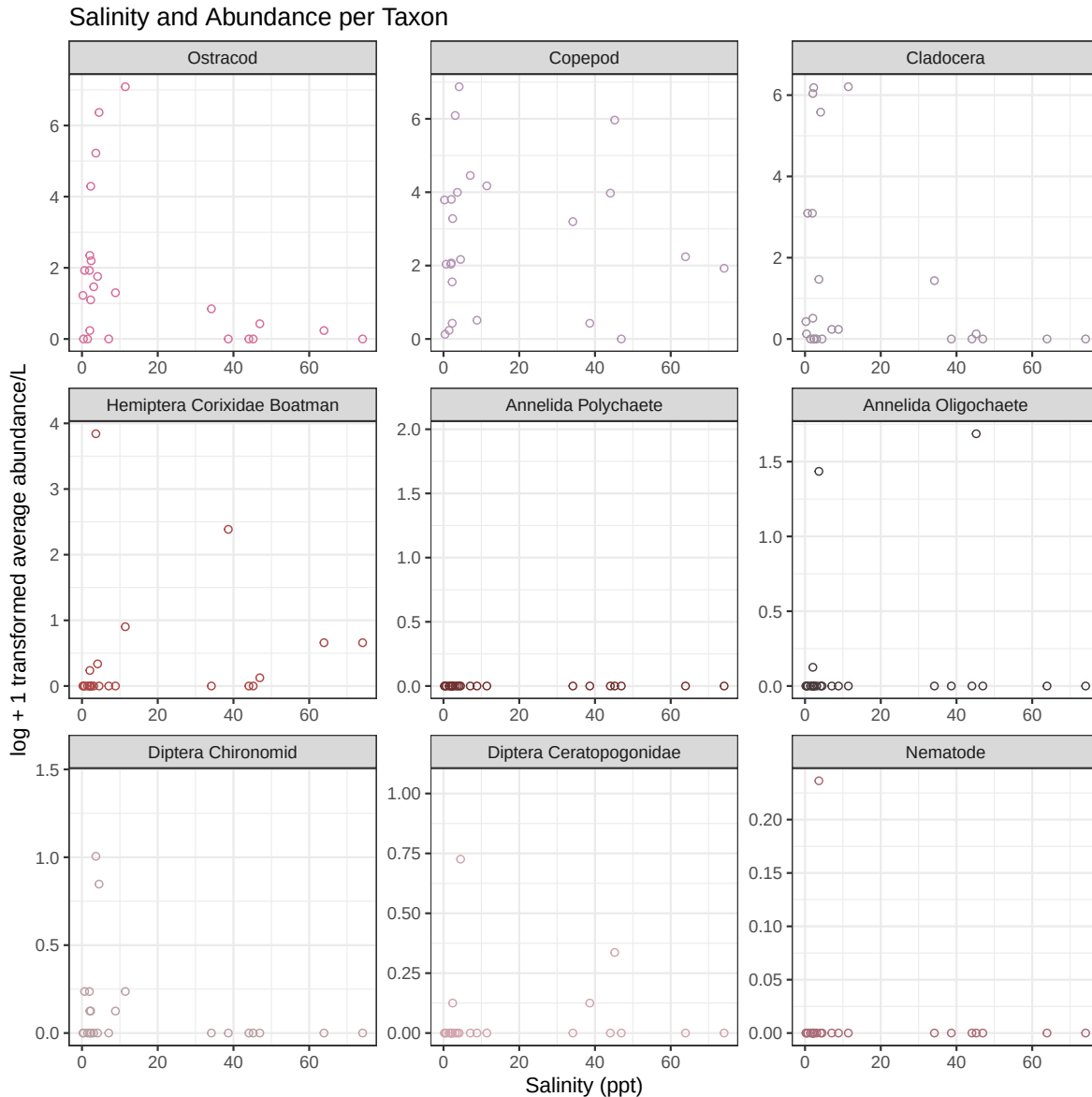
# labeling axes and title
labs(title = "Salinity and Abundance per Taxon",
      x = "Salinity (ppt)",
      y = "log + 1 transformed average abundance/L") +

# changed theme
theme_bw() +

# remove legend
theme(legend.position = "none")

# display plot
abund_salinity_plot

```



d. Interpret your figure

Across all sites, most of the data points are clustered in between a salinity of 0-10 ppt. Additionally, at all sites, there is a small cluster of data points around 40 ppt and 65 ppt, but not much in between 10-40 ppt and 50-60 ppt. For all taxa except for Annelida Polychaete and Annelida Oligochaete, the highest abundance/L points lie in between this salinity range of 0-10 ppt. Ostracod, Copepod, and Cladocera have the most data points that range across average

abundances per liter. Thus, it is reasonable to conclude that the lower salinity contributes to higher taxa abundance/L.